

Harsh Kumar Shrestha

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SUMMARY

Driven Full-Stack Developer skilled in Python, Node.js, and React.js. Experienced in building scalable microservices, gRPC APIs, and integrating Agentic AI solutions. Proficient in PostgreSQL, Redis, Docker, and system design, with a strong focus on writing clean, testable code.

EDUCATION

Birla Institute of Technology, Mesra

August 2023 - Expected May 2028

Integrated M.Sc. - Mathematics & Computing (GPA : 9.10 / 10.00)

Ranchi, Jharkhand

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Operating Systems, System Design, Web Technologies

PROJECTS

Agentic AI Customer Support Platform

React.js, Node.js, Python, LLMs

- Built a full-stack Agentic AI platform using React.js and Node.js, automating 70% of user queries via LLMs.
- Engineered Python backend microservices to process LLM prompts, reducing response latency by 50ms using Redis caching.
- Developed RESTful and gRPC APIs for frontend-backend communication, ensuring 99.9% uptime and persistent data in PostgreSQL.

Real-Time Event-Driven Collaboration Tool

Node.js, WebSockets, Redis, Docker

- Built an event-driven collaboration tool supporting 10,000 concurrent users using Node.js, React.js, and WebSockets.
- Designed microservices architecture with gRPC APIs, improving inter-service communication speed by 40% compared to REST.
- Integrated Redis for session management and PostgreSQL for data storage, ensuring scalable and testable code.

Cloud-Native E-Commerce Backend

Python, PostgreSQL, Docker, CI/CD

- Developed a scalable Python backend processing 5,000+ daily transactions, utilizing PostgreSQL for data persistence.
- Containerized backend services using Docker and automated deployments via CI/CD pipelines, reducing release time by 30%.
- Optimized database load by 40% by implementing Redis caching and enterprise security protocols.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript/TypeScript, Java, C++, SQL

Full-Stack & Frameworks: React.js, Node.js, Express.js, HTML/CSS

Backend & Architecture: RESTful APIs, gRPC, Event-Driven Systems, Microservices, System Design

Databases & Caching: PostgreSQL, MongoDB, Redis, WebSockets

AI & Cloud Infrastructure: Agentic AI, LLMs, Docker, CI/CD, Enterprise Security

Computer Science: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Unit Testing

CERTIFICATIONS

Linux & SQL: Google

Data Analysis with Python: Online Certification